

All fixed order asymptotics for distinct parabolic double cosets

Research manuscript with reproducible computations

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Abstract

Let p_n count the distinct subsets of the symmetric group S_n that arise as double cosets of standard parabolic subgroups. Starting from Browning's exact enumeration, we prove a Poincaré expansion to every fixed inverse power of n . We explicitly determine the first four correction polynomials. In particular, the first correction proves Browning's higher-order conjecture with $c = 0.10819705289392157158\dots$. The leading equivalent is Browning's previously established theorem. The proof supplies uniform generalized-Fubini pole extraction, summable bounds for the signed inner defects, and a joint bound for all omitted long-cycle configurations in the outer Stirling transform. We also give a finite exact coefficient algorithm, smooth asymptotic inverse models, Newton error estimates, and qualified brackets for the integer threshold. Exact computations and actual group-coset enumeration provide independent diagnostics. No claim of convergence, optimal truncation, exponential improvement, or exhaustive literature novelty is made.

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1 The counting problem and the main result

For $n \geq 1$, let S_n be the symmetric group on $\{1, \dots, n\}$ with adjacent transpositions $s_i = (i, i+1)$. For $I \subseteq \{1, \dots, n-1\}$, write $W_I = \langle s_i : i \in I \rangle$. Define

$$p_n = |\{W_I w W_J : I, J \subseteq \{1, \dots, n-1\}, w \in S_n\}|,$$

where the outer set identifies equal subsets of S_n , even if they have different presentations. Set $p_0 = 1$. These are the numbers recorded as OEIS A260700 [2], beginning

$$p_1, p_2, p_3, p_4, p_5, p_6 = 1, 3, 19, 167, 1791, 22715.$$

For example, in $S_2 = \{e, s_1\}$ the distinct subsets are $\{e\}$, $\{s_1\}$ and S_2 . This distinction is essential: summing double-quotient sizes over separately chosen subgroups retains duplicate presentations. Arbitrary contingency tables with no zero rows or columns form the different sequence A120733, whose value at $n = 2$ is 5 [3].

Throughout, put

$$\rho = \log 2, \quad K = \frac{e^{-\rho^2/2}}{4\rho^2} = 0.40922300504774271232 \dots \quad (1)$$

Browning proved $p_n \sim Kn! \rho^{-2n}$ [1]. His higher-order conjecture asks whether

$$\frac{p_n \rho^{2n}}{n!} = K - \frac{c}{n} + O(n^{-2}) \quad \text{for some } c > 0. \quad (2)$$

It is Conjecture 5.1 in the arXiv version and Conjecture 41 in the journal version, Section 5.1 in both. The following result settles that stated conjecture within this manuscript's proof.

Theorem 1.1. *There are explicitly computable polynomials $B_m(r) \in \mathbb{Q}[r]$, independent of the chosen truncation order, such that for every fixed integer $M \geq 0$,*

$$p_n = Kn! \rho^{-2n} \left(\sum_{m=0}^M \frac{B_m(\rho)}{n^m} + O_M(n^{-M-1}) \right). \quad (3)$$

The first three polynomials are $B_0(r) = 1$ and

$$B_1(r) = \frac{r^2}{2} - r^3 - \frac{11r^4}{12} + \frac{r^5}{4}, \quad (4)$$

$$B_2(r) = \frac{r^4}{288} \left(9r^6 - 66r^5 + 25r^4 + 552r^3 - 360r^2 - 720r + 264 \right). \quad (5)$$

Consequently (2) holds with

$$c = -KB_1(\rho) = 0.108197052893921571585327453732 \dots > 0. \quad (6)$$

The quantifier “every fixed M ” matters. The theorem does not assert any bound uniform in M , convergence of the infinite series, or a valid choice of M growing with n . Its contribution beyond the cited leading term consists of explicit corrections, justified uniform remainders at all fixed orders, and the inverse consequences developed below. The exact enumerative identities remain attributed to Browning; the analytic proof here begins from them.

A bounded primary-source search through 2 October 2026 did not locate a later resolution of this precise correction conjecture. Section 9 records the scope and neighboring problems. This observation is not a certification of universal novelty or a claim of external peer review.

[write] Added 2 October 2026, batch 77. This report is a single manuscript, printed in full: manuscript 17 of batch 77 of ProveIt’s incoming-reports intake, delivered as the archive `double-coset-report.zip` (wrapper directory `double-coset-report/`, main file already named `article.tex`) in commit 096ee7b87, and placed in commit d0e6008d9 in the research-report collection, among the OEIS-sequence asymptotics reports. The manuscript names no ProveIt commit and describes no search of the repository, so it has no pin; its author line (above) names neither an author nor a tool. The bounded search described in the preceding paragraph is the manuscript’s own literature search. The intake’s own untruncated search of the repository (the research-report collection, the transseries volumes and `0eis/`) found no other treatment of A260700, of Browning’s conjecture, or of parabolic double cosets. Nothing in this report is formalized in Lean or Rocq, and placement in the collection confers no formal status on any statement here. The writing step prefixed every label with `pd:`, added these **[write]** notes and three references [10, 11, 12], and set the bibliography ragged-right; no statement, proof or number of the manuscript was altered.

A neighbouring report. The same batch delivered (as manuscript 02) a study of the full-support elements of the Weyl groups of types B and D (OEIS A260952, A109253, A112225), now the collection report `a260952-full-support-signed-permutations` [10]. Both reports count objects defined by the standard parabolic subgroups of a Coxeter group — here the distinct double cosets $W_I w W_J$ in type A , there the elements lying in no proper standard parabolic subgroup in types B and D — and both reach a factorial scale through a Stirling transform and invert it with a Lambert W seed. Their theorems, generating functions and scales are different ($\rho = \log 2$ and $n!\rho^{-2n}$ here; $\tau = \log 3$ and $m!/(\log 3)^{m+1}$ there), and neither report uses the other. In particular [10] does not answer question 5 of Section 9 (distinct parabolic double cosets in another Coxeter family): it counts elements, not double cosets.

Notation. Each symbol is printed as delivered; several letters have two meanings in different sections, and some recur with other meanings in [10]. The table fixes them, with the tempting false readings.

symbol	meaning here (and what it is <i>not</i>)
p_n, q_n	p_n is A260700, q_n Browning's count of pairs of weak orders (10). In [10], $p_m(q)$ are <i>asymptotic coefficients</i> and q is a sign weight.
$B_m(r)$	the correction polynomials of Theorem 1.1, evaluated at $r = \rho$. Not Bernoulli polynomials, and not the counts $B_n(q)$ of signed permutations in [10].
ρ, K, c	$\rho = \log 2$, $K = e^{-\rho^2/2}/(4\rho^2)$ (1), $c = -KB_1(\rho) > 0$ (6); elsewhere c is the number of even blocks, a summation index (Sections 2–6). The analogue of ρ in [10] is $\tau = \log 3$; its K is a different carrier constant, and its c_n count indecomposable permutations.
$S_a(x)$	the Faulhaber power sum $\sum_{i < x} i^a$ (Section 2); not a Stirling number, which is always written with brackets or braces.
$H_j(N), Q(n)$	normalized Fubini counts and normalized q_n (13); not harmonic numbers, and $Q(n)$ is not the field $\mathbb{Q}$.
A	the germ $A(x)$ of (15) (Section 3); the constant $A = \frac{1}{2} \log(2\pi x_0) + \log K$ (Section 7).
D, D_m	$D_m(r)$ are the inner polynomials of Proposition 4.1; $\bar{D} = 1 + w$ in (44).
$a, a_s(j)$	$a_s(j)$ are the coefficients of $A(x)^{j+1}$; a is the cycle excess of (30), and an exponent index elsewhere.
L, L_n	L_n is the normalization (13); L the weighted cycle count of (30).
$T, T_a, \bar{T}_\mu$	the cutoff $T = C_M \log n$; the Touchard polynomials (27); the moment functional of Section 6.
R, R_n	$R(n, c, d)$ of (19); the contour radius R in the proof of Lemma 3.1; $R_n = p_n/(Kn!\rho^{-2n})$ in Section 8.
C, C_d, C_1, C_M	absolute or order-dependent constants C, C_0, C_R, C_M ; the coefficients $C_d(c)$ of $J(z)^c$; $C_1 = \frac{1}{12} + B_1(\rho)$ in Section 7.
μ, λ	$\mu = -\rho^2$ in the signed moment (a “signed Poisson average”, not a probability); $\lambda = \rho^2/2$ in the outer sum.
t, w, x	t is a summation index ($t = 2c + d$) in Sections 2–4 and the formal variable $1/n$ in Section 6; w is a permutation in Section 1 and the Lambert value in (44); x is the germ variable in Section 3 and a real index in Section 7.
$E_m, \mathbb{E}$	the scaled residuals $E_m(n)$ of Section 8; the binomial expectation $\mathbb{E}$.

2 The exact enumeration and normalization

Use the falling factorial $(x)_j = x(x-1)\cdots(x-j+1)$, with $(x)_0 = 1$. Write $\begin{bmatrix} n \\ m \end{bmatrix}$ and $\{n\}_m$ for unsigned Stirling numbers of the first and second kinds. For $a \geq 1$, let

$$S_a(x) = \sum_{i=0}^{x-1} i^a$$

mean the Faulhaber polynomial, so this notation also makes sense at noninteger x .

A weak order is an ordered set partition. Let $f_{N,j}$ count weak orders of an N -element set in which j specified elements must each occupy a singleton block. Then

$$f_{N,j} = \sum_{\ell=j}^N \ell! \begin{Bmatrix} N-j \\ \ell-j \end{Bmatrix}, \quad (7)$$

$$f_{N,j} = j!(N-j)![z^{N-j}](2 - e^z)^{-j-1}. \quad (8)$$

For (7), first partition the other $N-j$ elements into $\ell-j$ blocks and then order all ℓ blocks. To obtain (8), use the exponential generating function of set partitions and

$$\sum_{a \geq 0} \frac{(a+j)!}{a!} u^a = j!(1-u)^{-j-1}, \quad u = e^z - 1.$$

In particular $0 \leq f_{N,j} \leq f_{N,0}$.

The exact identities we use are Browning's Theorem 3.18, Proposition 3.19 and Section 4.1 in arXiv numbering [1]:

$$p_n = \frac{1}{n!} \sum_{m=0}^n \begin{bmatrix} n \\ m \end{bmatrix} q_m, \quad (9)$$

$$q_n = \sum_{\substack{c,t \geq 0 \\ 2c \leq t \leq n}} \binom{n}{t} h_{t,c} \sum_{j=0}^c \frac{f_{n-t+c,j} f_{n-t+c,c-j}}{j!(c-j)!}, \quad (10)$$

$$h_{t,c} = (-2)^c t! [z^t] (\cosh z - 1)^c. \quad (11)$$

The last identity follows from Browning's interpretation of $(-1)^c h_{t,c}/2^c$ as ordered partitions into c even, nonempty blocks. His q_n counts pairs of weak orders without consecutive embeddings; hence

$$0 \leq q_n \leq f_{n,0}^2. \quad (12)$$

We use the count and this bound, rather than assume that cancellation in (10) preserves a termwise sign. The identities include $n = 0$ with the usual empty-object conventions.

Normalize by

$$L_n = \frac{n!^2}{4\rho^{2n+2}}, \quad Q(n) = \frac{q_n}{L_n}, \quad H_j(N) = \frac{f_{N,j}}{N!/(2^{j+1}\rho^{N+1})}. \quad (13)$$

The simple pole of $(2 - e^z)^{-1}$ at $z = \rho$ gives, after adjusting finitely many indices,

$$f_{N,0} \leq CN! \rho^{-N-1} \quad (N \geq 0) \quad (14)$$

for an absolute constant C . This also follows from the next lemma with $j = 0$.

3 Uniform generalized Fubini asymptotics

The extraction must remain uniform for logarithmically growing j . A fixed- j asymptotic would not justify the later sums. Define the analytic germ

$$A(x) = \frac{\rho x}{1 - e^{-\rho x}}, \quad a_s(j) = [x^s] A(x)^{j+1}. \quad (15)$$

Here $A(0) = 1$, and

$$A(x) = 1 + \frac{\rho x}{2} + \frac{\rho^2 x^2}{12} + O(x^4).$$

For fixed s , the coefficient $a_s(j)$ is a polynomial in j and ρ with rational coefficients and degree at most s in j . Indeed, expand $(1 + (A - 1))^{j+1}$, retaining only the finitely many powers up to s .

Lemma 3.1. *Fix $C_0 > 0$. Uniformly for integers $0 \leq j \leq C_0 \log n$ and $N = n + O(\log n)$,*

$$H_j(N) = \sum_{s=0}^j a_s(j) \frac{(j)_s}{(N)_s} + O\left(e^{-\eta n + O((\log n)^2)}\right) \quad (16)$$

for some $\eta > 0$. For every fixed M , the same sum truncated at $s \leq M$ has error

$$O_M\left(n^{-M-1}(j+1)^{2M+2}\right) + O\left(e^{-\eta n + O((\log n)^2)}\right). \quad (17)$$

In particular, $H_j(N) = O(1)$ in this range and

$$H_j(N) = 1 + \frac{\rho j(j+1)}{2N} + O\left(n^{-2}(j+1)^4\right). \quad (18)$$

Proof. Choose a fixed radius R with

$$\rho < R < \sqrt{\rho^2 + 4\pi^2}.$$

The poles of $F(z) = (2 - e^z)^{-1}$ are $\rho + 2\pi i\ell$, $\ell \in \mathbb{Z}$, so only ρ lies inside $|z| = R$. Near ρ , with $x = 1 - z/\rho$,

$$F(z)^{j+1} = (2\rho)^{-j-1} x^{-j-1} A(x)^{j+1}.$$

Its principal part is therefore

$$(2\rho)^{-j-1} \sum_{s=0}^j a_s(j) (1 - z/\rho)^{-j-1+s}.$$

Deform the coefficient contour for $[z^{N-j}]F(z)^{j+1}$ to $|z| = R$. Extracting the residue contribution from this principal part and multiplying by $j!(N-j)!$ gives the normalized coefficient ratio

$$\frac{\binom{N-s}{j-s}}{\binom{N}{j}} = \frac{(j)_s}{(N)_s}.$$

On the outer circle, the coefficient integral is at most $R^{-N+j} C_R^{j+1}$ in absolute value. After the normalization in (13), all remaining factorial and exponential factors contribute at most $\exp(O(j \log n + j \log(j+1)))$. Since $(\rho/R)^N$ decays exponentially, this proves (16). This argument uses contour deformation; no global analytic subtraction of the principal part is needed.

For the finite principal-part tail, take a small fixed $\delta > 0$. On $|x| = \delta/(j+1)$, the bound $\log A(x) = O(x)$ gives $|A(x)^{j+1}| \leq C$. Cauchy's bound yields

$$|a_s(j)| \leq C(C(j+1))^s.$$

For $s \leq j$ and large n ,

$$\left| \frac{(j)_s}{(N)_s} \right| \leq \left(\frac{j}{N-j} \right)^s.$$

The terms $s \geq M+1$ are thus bounded by a geometric tail with ratio $O((j+1)^2/n)$, proving (17). If $j < M$, the extra falling-factorial terms vanish. Every retained reciprocal falling factorial can be expanded to a fixed order with a polynomial bound in j and the $O(\log n)$ shift of N . Finally $a_1(j) = \rho(j+1)/2$, giving (18). $\square$

4 The signed inner defect expansion

Set

$$J(z) = \frac{2(\cosh z - 1)}{z^2}, \quad C_d(c) = [z^d]J(z)^c, \quad R(n, c, d) = n^d \frac{(n)_{2c+d}}{(n)_{c+d}^2}. \quad (19)$$

The removable value is $J(0) = 1$. All coefficients of J are nonnegative, $C_d(c) = 0$ for odd d , $C_0(c) = 1$ and $C_2(c) = c/12$. For each fixed even d , $C_d(c)$ is a rational polynomial in c of degree at most $d/2$.

With $d = t - 2c$, formula (11) becomes

$$h_{2c+d,c} = (-1)^c (2c + d)! C_d(c).$$

Substitution into (10), followed by the normalizations, gives the exact identity

$$Q(n) = \sum_{\substack{c,d \geq 0 \\ 2c+d \leq n}} \frac{(-\rho^2)^c}{c!} \frac{\rho^{2d} C_d(c)}{n^d} \mathbb{E}_{j \sim \text{Bin}(c, 1/2)} [R(n, c, d) H_j(n - c - d) H_{c-j}(n - c - d)]. \quad (20)$$

The $2^{-c} \binom{c}{j}$ in the binomial expectation is precisely the factor left by the two normalized Fubini quantities and $j!(c - j)!$.

4.1 A positive bound for all large signed defects

We first discard large $t = 2c + d$ using the original exact sum, before invoking any local expansion. If $2c \leq t \leq n$, then

$$\binom{n}{t} \left(\frac{(n - t + c)!}{n!} \right)^2 \leq \frac{1}{t!}. \quad (21)$$

Indeed, $2(n - t + c) \leq (n - t) + n$, and monotonicity and log-convexity of factorials give $(n - t + c)!^2 \leq (n - t)!n!$. Using $f_{N,j} \leq f_{N,0}$, (14), and $0 < \rho < 1$, the absolute normalized contribution of a given (t, c) in (10) is at most

$$C \frac{2^c |h_{t,c}|}{c!t!}. \quad (22)$$

For every fixed $u > 0$, the positive series has the entire generating function

$$\sum_{t,c \geq 0} \frac{2^c |h_{t,c}|}{c!t!} u^t = \exp(4(\cosh u - 1)) < \infty. \quad (23)$$

Hence for $u > 1$ the sum over $t > T$ is $O_u(u^{-T})$. Given M , choose $T = C_M \log n$ sufficiently large to make this $O_M(n^{-M-2})$. Derivatives of the entire generating function show that every fixed polynomial weight in t, c is summable as well. This is an absolute estimate for a signed sum; dominated convergence alone would not supply the required all-orders error.

4.2 Local expansion and the remaining defect tail

On $t \leq T$ the convergent logarithmic expansion is

$$\log R(n, c, d) = - \sum_{a \geq 1} \frac{S_a(2c + d) - 2S_a(c + d)}{an^a}. \quad (24)$$

It follows by expanding $\log(1 - i/n)$ in each falling factorial. Uniformly in this range, $R(n, c, d) \leq \exp(Ct^2/n) = O(1)$. Truncating (24) and exponentiating to a fixed order gives an error bounded by n^{-M-1} times a fixed polynomial in t , times $\exp(Ct^2/n)$.

Lemma 3.1 also gives $H_j(n - c - d) = O(1)$ uniformly here. Terms with $d > M$ in (20) are bounded by $O_M(n^{-M-1})$, because

$$\sum_{c,d \geq 0} \frac{\rho^{2c+2d}}{c!} C_d(c) = \exp(\rho^2 J(\rho^2)) < \infty. \quad (25)$$

The argument of J is ρ^2 . More generally

$$\sum_{c,d \geq 0} \frac{\rho^{2c+2d}}{c!} C_d(c) v^c u^d = \exp(v\rho^2 J(u\rho^2))$$

is entire in both variables. Thus polynomial weights in c, d are summable.

For each even $d \leq M$, substitute the finite expansions from Lemma 3.1 and (24) into (20). Every coefficient is a polynomial in c, j, ρ , and the error has a summable polynomial weight times n^{-M-1} . At each retained d , extending the coefficient sums from $c \leq (T - d)/2$ to all $c \geq 0$ adds a superfactorially decaying tail: a polynomial times $\rho^{2c}/c!$ beyond a constant multiple of $\log n$ is smaller than every fixed inverse power of n .

Proposition 4.1. *For every fixed M ,*

$$Q(n) = e^{-\rho^2} \left(\sum_{m=0}^M D_m(\rho) n^{-m} + O_M(n^{-M-1}) \right), \quad (26)$$

where $D_m(r) \in \mathbb{Q}[r]$ and $D_0 = 1$.

Proof. All error estimates have just been established. To evaluate coefficients, convert powers of j to falling factorials and use

$$\mathbb{E}_{j \sim \text{Bin}(c, 1/2)}(j)_a = 2^{-a}(c)_a.$$

The remaining polynomial in c is summed against $\mu^c/c!$ with $\mu = -\rho^2$. For a nonnegative integer a , the Touchard identity is

$$e^{-\mu} \sum_{c \geq 0} \frac{\mu^c c^a}{c!} = T_a(\mu) = \sum_{b=0}^a \left\{ \begin{matrix} a \\ b \end{matrix} \right\} \mu^b. \quad (27)$$

It is an entire identity in μ , so it also holds at the negative argument. Absolute summability justifies its use; a “signed Poisson average” is an algebraic functional, not a probability distribution. Equation (26) follows. $\square$

5 The outer Stirling transform

The exact normalized form of (9) is

$$\frac{p_n}{n!/(4\rho^{2n+2})} = \sum_{k=0}^n \rho^{2k} \frac{\begin{bmatrix} n \\ n-k \end{bmatrix}}{(n)_k^2} Q(n-k). \quad (28)$$

We need both a global tail estimate and a local expansion that handles all omitted cycle types together.

5.1 Global tail of the outer defect

The Stirling number is an elementary symmetric polynomial in $0, 1, \dots, n-1$. Consequently

$$\begin{bmatrix} n \\ n-k \end{bmatrix} \leq \frac{n^{2k}}{2^k k!}.$$

By (12) and (14), $Q(m)$ is bounded uniformly in m . For $n \geq k \geq 1$, the product

$$\frac{n^{2k}(n-k)!^2}{n!^2} = \prod_{i=0}^{k-1} \left(\frac{n}{n-i} \right)^2$$

is decreasing in n and at most $k^{2k}/(k!)^2$. Thus the absolute k th summand of (28) is bounded by

$$C \frac{\rho^{2k} k^{2k}}{2^k (k!)^3}, \quad (29)$$

with the $k=0$ factor interpreted as 1. This sequence has every exponential moment: multiplying by u^k for any fixed $u > 0$ leaves a convergent series, by Stirling's formula. As in Section 4, terms $k > C_M \log n$ can therefore be discarded at any prescribed fixed inverse-power order.

5.2 Cycle configurations and their joint remainder

For $k \leq C_M \log n$, choose multiplicities m_ℓ of cycles of lengths $\ell \geq 3$, and put

$$a = \sum_{\ell \geq 3} (\ell-2)m_\ell, \quad L = \sum_{\ell \geq 3} (\ell-1)m_\ell, \quad m_2 = k - L. \quad (30)$$

The total number of moved labels is $2k - a$. The usual permutation cycle count yields

$$\frac{2^k k! \begin{bmatrix} n \\ n-k \end{bmatrix}}{(n)_k^2} = \sum_{\substack{(m_\ell)_{\ell \geq 3} \\ L \leq k}} \frac{2^L (k)_L}{\prod_{\ell \geq 3} \ell^{m_\ell} m_\ell!} \cdot \frac{(n)_{2k-a}}{(n)_k^2}. \quad (31)$$

Here $2k < n$ for large n , so every listed support is feasible.

For a fixed configuration with $a \leq M$, the ratio is

$$\frac{(n)_{2k-a}}{(n)_k^2} = n^{-a} \exp \left(- \sum_{b \geq 1} \frac{S_b(2k-a) - 2S_b(k)}{bn^b} \right). \quad (32)$$

There are finitely many such configurations, all with $\ell \leq M+2$. Their fixed-order expansions have polynomial coefficients in k . Although a polynomial formula may be extended to all integers $k \geq 0$, the factor $(k)_L$ vanishes whenever $k < L$, so that this extension inserts no false contribution.

To bound all configurations with $a > M$ simultaneously, the factorial ratio in (31) is at most

$$n^{-a} \exp(Ck^2/n),$$

and the combinatorial prefactor is at most

$$\prod_{\ell \geq 3} \frac{(2^{\ell-1} k^{\ell-1} / \ell)^{m_\ell}}{m_\ell!}.$$

Use $\ell - 1 \leq 2(\ell - 2)$ and $2^{\ell-1} \leq 4^{\ell-2}$. With $x = 4k^2/n$, the sum at a given total excess a is bounded by the degree- a term in

$$\exp \left(\sum_{h \geq 1} \frac{x^h}{h+2} \right). \quad (33)$$

This power series has nonnegative coefficients and is analytic for $|x| < 1$. For $0 \leq x \leq 1/4$, its tail after degree M is $O_M(x^{M+1})$. Therefore the omitted part of (31) is

$$O_M \left(n^{-M-1} (k+1)^{2M+2} e^{Ck^2/n} \right). \quad (34)$$

This joint bound is necessary; expansions of each fixed cycle type alone would not control the infinite family of omitted types.

It follows that there are polynomials $U_m(k) \in \mathbb{Q}[k]$ such that, uniformly in the logarithmic range,

$$\frac{2^k k! \begin{bmatrix} n \\ n-k \end{bmatrix}}{(n)_k^2} = \sum_{m=0}^M U_m(k) n^{-m} + O_M \left(n^{-M-1} (1+k)^{A_M} \right) \quad (35)$$

for some finite exponent A_M . The first three are

$$U_0(k) = 1, \quad U_1(k) = \frac{k(k-4)}{3}, \quad U_2(k) = \frac{k^2(k-5)(k-1)}{18}. \quad (36)$$

5.3 Completion of the main proof

Insert (26) and (35) into (28), and expand

$$(n-k)^{-i} = n^{-i} (1 - k/n)^{-i}$$

to the required fixed order. All resulting error bounds are n^{-M-1} times polynomial weights in k , summable against $\lambda^k/k!$ with $\lambda = \rho^2/2$. Extending the coefficient sums to all $k \geq 0$ adds another superfactorial tail. Their moments are $e^\lambda T_a(\lambda)$. The prefactors combine as

$$e^{-\rho^2} e^{\rho^2/2} = e^{-\rho^2/2},$$

giving (3) and $B_m \in \mathbb{Q}[r]$. Expansions at successive truncation orders are compatible; alternatively uniqueness of a Poincaré expansion shows the polynomials evaluated at ρ do not depend on the chosen cutoff. The finite algebraic construction below chooses a single compatible polynomial $B_m(r)$ at every order. This proves the all-orders part of Theorem 1.1.

6 Finite coefficient computation and the first corrections

The proof is constructive. Here is a precise finite algorithm using the formal variable $t = 1/n$ and polynomial arithmetic over $\mathbb{Q}[r]$. All series in this section are truncated modulo t^{M+1} .

For a polynomial $P(c)$ define the moment functional

$$\mathcal{T}_\mu(P) = \sum_{a \geq 0} [c^a] P(c) T_a(\mu).$$

The sum is finite. Define $\mathcal{B}_c$ on polynomials in j by

$$\mathcal{B}_c((j)_a) = 2^{-a}(c)_a.$$

For each even $0 \leq d \leq M$, compute $C_d(c)$ from J and form

$$\begin{aligned} \mathcal{R}_{c,d}(t) &= \exp \left(- \sum_{a=1}^M \frac{S_a(2c+d) - 2S_a(c+d)}{a} t^a \right), \\ \mathcal{H}_j(c+d;t) &= \sum_{s=0}^M a_s(j)(j)_s t^s \prod_{i=0}^{s-1} (1 - (c+d+i)t)^{-1}. \end{aligned}$$

Then the inner coefficients satisfy

$$\sum_{m=0}^M D_m(r) t^m = \mathcal{T}_{-r^2} \left[\sum_{\substack{0 \leq d \leq M \\ d \text{ even}}} r^{2d} C_d(c) t^d \mathcal{B}_c(\mathcal{R}_{c,d}(t)) \mathcal{H}_j(c+d;t) \mathcal{H}_{c-j}(c+d;t) \right]. \quad (37)$$

Here $a_s(j)$ is computed with the formal parameter r in place of ρ .

For the outer coefficients, enumerate the finitely many configurations with $a \leq M$ in (30) and compute

$$\sum_{m=0}^M U_m(k) t^m = \sum_{a \leq M} \frac{2^L(k)_L t^a}{\prod_{\ell \geq 3} \ell^{m_\ell} m_\ell!} \exp \left(- \sum_{b=1}^M \frac{S_b(2k-a) - 2S_b(k)}{b} t^b \right), \quad (38)$$

where the sum is over those configurations. Finally,

$$B_m(r) = \mathcal{T}_{r^2/2} \left([t^m] \left\{ \left(\sum_{a=0}^M U_a(k) t^a \right) \left(\sum_{i=0}^M D_i(r) t^i (1-kt)^{-i} \right) \right\} \right), \quad (39)$$

now interpreting the moment functional on the variable k . Equations (37)–(39) are an arbitrary-fixed-order algorithm, not a fit to numerical sequence values.

6.1 A direct derivation of the first correction

Only $d = 0$ is needed through order n^{-1} . From (24),

$$R(n, c, 0) = 1 - \frac{c^2}{n} + O(n^{-2} \text{poly}(c)).$$

The binomial average of the two Fubini corrections is

$$\frac{\rho}{2} \mathbb{E}(j(j+1) + (c-j)(c-j+1)) = \frac{\rho(c^2 + 3c)}{4}.$$

Taking signed moments at $\mu = -\rho^2$ gives

$$\begin{aligned} D_1(\rho) &= -(\mu^2 + \mu) + \frac{\rho}{4}(\mu^2 + 4\mu) \\ &= \rho^2 - \rho^3 - \rho^4 + \frac{\rho^5}{4}. \end{aligned} \quad (40)$$

For the outer term, the all-transposition configuration contributes $-k^2$ at first order, and one 3-cycle contributes $4k(k-1)/3$. Their sum is $U_1(k) = k(k-4)/3$. At $\lambda = \rho^2/2$,

$$\mathcal{T}_\lambda(U_1) = \frac{\lambda^2 - 3\lambda}{3} = \frac{\rho^4}{12} - \frac{\rho^2}{2}.$$

Adding this to (40) gives (4). Finally $2/3 < \rho < 1$ and

$$\frac{B_1(\rho)}{\rho^2} = \frac{1}{2} - \rho - \rho^2 \left(\frac{11}{12} - \frac{\rho}{4} \right) < \frac{1}{2} - \frac{2}{3} < 0.$$

For completeness, $\log 2 > 2/3$ follows by integrating the strictly convex function $1/x$ on $[1, 2]$ and comparing with its midpoint value; $\log 2 < 1$ follows from $1/x < 1$ for $1 < x \leq 2$. Thus $c > 0$ is proved without a numerical sign test.

6.2 Second and higher corrections

At second order the first nonzero extra inner defect is $d = 2$, contributing $r^4 c/12$ before the signed moment. The exact algorithm gives

$$D_2(r) = \frac{r^4}{96} \left(3r^6 - 24r^5 + 16r^4 + 192r^3 - 128r^2 - 240r + 96 \right).$$

Including the shift from $Q(n-k)$, the second outer expression is

$$D_2 + D_1(k + U_1(k)) + U_2(k).$$

Applying $\mathcal{T}_{r^2/2}$ gives (5). Appendix A records B_3, B_4 ; all D_m, U_m, B_m through order four are in the accompanying exact JSON data. Numerically,

$$\begin{aligned} B_1(\rho) &= -0.26439631095836993468\dots, \\ B_2(\rho) &= -0.18273691798917761621\dots, \\ B_3(\rho) &= -0.32435913595799104018\dots, \\ B_4(\rho) &= -0.69063295859309244992\dots \end{aligned}$$

These first signs do not establish a sign pattern for all coefficients.

7 Smooth inverse models and integer thresholds

For each fixed M , let

$$P_M(t) = \sum_{m=0}^M B_m(\rho) t^m, \quad F_M(x) = K\Gamma(x+1)\rho^{-2x}P_M(1/x). \quad (41)$$

These functions are positive for all sufficiently large real x , and

$$\frac{d}{dx} \log F_M(x) = \psi(x+1) - 2\log \rho - \frac{P'_M(1/x)}{x^2 P_M(1/x)} = \log(x/\rho^2) + O(1/x) > 0. \quad (42)$$

Here ψ is the digamma function. Let $x_M(y)$ denote their eventual increasing inverse. This definition makes no choice of a canonical smooth interpolation of the discrete sequence.

Proposition 7.1. *As $y \rightarrow \infty$,*

$$x_M(y) - x_{M+1}(y) = O_M\left(\frac{x_M(y)^{-M-1}}{\log x_M(y)}\right). \quad (43)$$

The same error scale holds between x_M and the inverse of any positive, eventually strictly increasing smooth interpolation that differs from $F_M(x)$ by a relative $O(x^{-M-1})$.

Proof. The two logarithms differ by $O(x^{-M-1})$. By (42), changing x near the model root changes $\log F_M$ at a rate comparable to $\log x$. Apply the mean value theorem to $\log F_M$ at the two roots. For the interpolation statement, monotonicity is an explicit hypothesis ensuring an eventual inverse; relative value closeness alone would not ensure it. $\square$

7.1 Lambert W seed and explicit corrections

Let $Y = \log y$ and use the positive real branch of the Lambert W function. Define

$$w = W\left(\frac{Y}{e\rho^2}\right), \quad x_0 = \frac{Y}{w}, \quad D = 1 + w, \quad (44)$$

so $x_0(\log x_0 - 1 - 2\log \rho) = Y$. Set

$$A = \frac{1}{2} \log(2\pi x_0) + \log K, \quad C_1 = \frac{1}{12} + B_1(\rho).$$

For every model retaining B_1 , that is $M \geq 1$, its inverse begins

$$x_M(y) = x_0 - \frac{A}{D} - \frac{A^2/(2D^2) - A/(2D) + C_1}{x_0 D} + O_M\left(\frac{1}{x_0^2 \log x_0}\right). \quad (45)$$

To verify this, write the logarithm of (41) using Stirling's expansion:

$$\log F_M(x) = x(\log x - 1 - 2\log \rho) + \frac{1}{2} \log(2\pi x) + \log K + \frac{C_1}{x} + O_M(x^{-2}).$$

The first two derivatives of the leading term at x_0 are D and $1/x_0$. Put $x = x_0 + \delta$. The first cancellation is $D\delta + A = 0$, giving $\delta_0 = -A/D = O(1)$. The next residual is

$$\frac{1}{x_0} \left(\frac{\delta_0^2}{2} + \frac{\delta_0}{2} + C_1 \right),$$

which gives the second correction displayed in (45). The remaining logarithmic error is $O_M(x_0^{-2})$; division by a derivative comparable to $\log x_0$ gives the stated inverse error.

Arbitrary fixed orders follow by retaining more of the forward coefficients, the Stirling series for $\log \Gamma$, and ordinary Taylor series. The next proposition is also an explicit all-fixed-order numerical inversion algorithm with an error estimate.

Proposition 7.2. *Let $G_M(x) = \log F_M(x) - Y$ and start $X_0 = x_0$. Define*

$$X_{s+1} = X_s - \frac{G_M(X_s)}{G'_M(X_s)}. \quad (46)$$

For every fixed M and s , these iterates are well defined for all sufficiently large Y , remain within an $O(1)$ neighborhood of x_0 , and satisfy

$$X_s - x_M(y) = O_{M,s}\left((x_0 \log x_0)^{-(2^s-1)}\right). \quad (47)$$

Proof. The root is $x_0 + O(1)$ by the preceding logarithmic expansion. In a fixed neighborhood containing it, $G'_M(x) \asymp \log x_0$ and $G''_M(x) = O(1/x_0)$. For $e_s = X_s - x_M$, Taylor's theorem gives

$$|e_{s+1}| \leq \frac{C|e_s|^2}{x_0 \log x_0}.$$

Starting with $e_0 = O(1)$, induction gives the exponent $2^s - 1$. It also keeps every iterate in the neighborhood for large Y . $\square$

Choosing $2^s - 1 \geq M + 1$ makes the numerical iteration error no larger than the forward-model uncertainty in (43). Expanding these same Newton iterates symbolically yields inverse expansions to any prescribed fixed order. An asymptotic existence statement is distinct from an effective finite- y certificate: this manuscript does not extract global numerical constants or starting points for these bounds.

7.2 The integer threshold requires a rounding margin

Define the actual sequence threshold

$$N(y) = \min\{n \geq 0 : p_n \geq y\}.$$

For $n \geq 1$, $p_{n+1} > p_n$. Indeed, embed each parabolic double coset of S_n into S_{n+1} as a subset fixing $n+1$, using the same generators and embedded representative. The images remain distinct standard parabolic double cosets. The singleton consisting of $(n, n+1)$ is an additional coset. Alternatively, eventual monotonicity follows from Theorem 1.1.

Corollary 7.3. *For each fixed M , there is a constant $C_M > 0$ such that for all sufficiently large y ,*

$$\lceil x_M(y) - \varepsilon_M(y) \rceil \leq N(y) \leq \lceil x_M(y) + \varepsilon_M(y) \rceil, \quad \varepsilon_M(y) = \frac{C_M x_M(y)^{-M-1}}{\log x_M(y)}. \quad (48)$$

Proof. Theorem 1.1 gives $\log p_n - \log F_M(n) = O_M(n^{-M-1})$. On a bounded neighborhood of $x_M(y)$, the derivative of $\log F_M$ is comparable to $\log x_M(y)$. Choose C_M strictly larger than the required error bound. Then every integer below $x_M - \varepsilon_M$ has $p_n < y$, and every integer at or above $x_M + \varepsilon_M$ has $p_n \geq y$. Monotonicity and the fact that finite initial indices cannot reach sufficiently large y complete the assertion. $\square$

If the two ceiling endpoints agree, their common value is the threshold. Without an effective value of C_M and starting point, (48) is an asymptotic theorem, not a ready finite-input rounding test. An unconditional claim $N(y) = \lceil x_M(y) \rceil$ would be false. The reproducible example $y = p_{25} + 1$ has true threshold 26 but $\lceil x_1(y) \rceil = 25$: a one-unit change in y is much smaller than the forward approximation error.

[write] **Added 2 October 2026, batch 77: an instance of repository results.** The smooth inverse of this section is an instance of the factorial-core inversion of ProveIt's transseries volumes [11, 12], and no novelty is claimed for the method. The phase is $\log(\Gamma(x+1)\rho^{-2x}) = x(\log x - 1 - 2\log \rho) + O(\log x)$, so the seed (44) is the exact solution of $X(\kappa \log X + d) = L$ with $\kappa = 1$, $d = -1 - 2\log \rho$ and $L = Y$ in `p0:prop:factorial-core` of [11]: there $v = W_0((L/\kappa)e^{d/\kappa}) = W_0(Y/(e\rho^2)) = w$ and $X = e^{v-d/\kappa} = e\rho^2 e^w = Y/w = x_0$. It is equally the unbalanced core `t2:eq:unbalanced-core` of [12] with $\kappa = 1$, $\lambda = -1 - 2\log \rho$ and $L = \log y$ in place of $\log(y/C)$. In the variable $s = \log X + d/\kappa$ that equation reads $s + \log s = \log(L/\kappa) + d/\kappa$, which is `p0:thm:lambert-core` with $a = b = 1$; since $b > 0$ its

branch rule selects W_0 , the positive real branch used here. The explicit corrections (45) and every further fixed order are the reversion about an admissible core, `p0:thm:core-reversion` of [11] (in [12], the scaled local reversion `t2:prop:scaled`); the linear-logarithmic specialization `p0:thm:lambert-centered` and the balanced inverse `t2:thm:balanced-inverse` do not apply directly, because the phase is $x \log x$ rather than $ax + b \log x$. The bracket (48) is the separation condition, part (2) of `p0:thm:staircase`, with $x_* = x_M(y)$ and $\eta = \varepsilon_M(y)$. Part (1) of that theorem (exact rounding $N = \lceil \nu \rceil$) needs an admissible interpolation of the sequence itself; F_M is not one, since $F_M(n) = p_n(1 + O(n^{-M-1}))$ only, and the counterexample $y = p_{25} + 1$ above is exactly such a failure. It stays as stated. The generic staircase arithmetic is formalized in Lean as `Fabius.staircase_ceil` and `Fabius.staircase_separation` in `Analysis/FabiusFunction/Lean/FabiusFunction/StaircaseInversion.lean`; those statements concern an arbitrary monotone function and give no formal status to anything in this section. Specific to this report are the forward expansion of Theorem 1.1 that feeds the inversion, Proposition 7.1, the Newton estimate of Proposition 7.2, the bracket of Corollary 7.3 for p_n , and the counterexample.

8 Exact and numerical checks

The accompanying programs use integer or exact rational arithmetic for enumeration and coefficient generation. Floating-point arithmetic is reserved for numerical evaluation and inverse diagnostics. These checks test implementation, normalization and model identification; the remainder proofs above do not depend on numerical agreement.

1. The finite exact formula (9)–(11) reproduces every OEIS b-file value for $1 \leq n \leq 400$. The package includes the locally recomputed p_n, q_n and the separately attributed numerical reference data.
2. Independent enumeration of actual subsets $W_I w W_J$ in S_n , by connected components under the left and right adjacent-transposition actions, gives 1, 3, 19, 167, 1791 for $1 \leq n \leq 5$. Presentations are deduplicated as sets of permutations. This checks that the finite formula is applied to the intended distinct-subset problem.
3. Exact cycle-count checks verify 259 cases with $n \leq 40$ and small defects k . The symbolic coefficient generator uses exact rational operations and reproduces all displayed B_m , $m \leq 4$.
4. Numerical residuals use 90-digit arithmetic; inverse checks use 150 digits, sufficient to distinguish p_{25} from $p_{25} + 1$. No floating-point fit is used to infer any coefficient.

For $R_n = p_n / (Kn! \rho^{-2n})$, define

$$E_m(n) = n^{m+1} \left(R_n - \sum_{i=0}^m B_i(\rho) n^{-i} \right).$$

The theorem predicts $E_m(n) \rightarrow B_{m+1}(\rho)$. The recorded values are:

n	$E_0(n)$	$E_1(n)$	$E_2(n)$	$E_3(n)$
100	-0.2662568242	-0.1860513214	-0.3314403386	-0.7081202633
200	-0.2653181919	-0.1843761950	-0.3278554103	-0.6992548747
400	-0.2648551914	-0.1835521590	-0.3260964214	-0.6949141602
Limit	-0.2643963110	-0.1827369180	-0.3243591360	-0.6906329586

For $y = p_{100}$, the first three Newton updates for F_4 reduce the seed error from approximately 0.4356 to $1.7600 \cdot 10^{-4}$, $2.8841 \cdot 10^{-11}$ and $7.7446 \cdot 10^{-25}$. This illustrates rapid model inversion, while the distinction between that model and the exact discrete threshold remains in force.

9 Literature boundary and further questions

The exact counting formula, weak-order framework and leading constant are from Browning [1]. The neighboring Stirling-transform approach to table enumeration also appears in Kotěšovec's work on A120733 [4]; neither Stirling transforms nor general singularity analysis is introduced here as a new technique.

The bounded search checked the sequence identifier, the paper title and correction-conjecture wording, later double-coset work, and Browning's current publication listing [9]. In particular, Schwob's later paper [5] studies prescribed subgroup pairs and aggregates including A321652 and A178718; it does not identify equal double-coset subsets across all presentations in the way required for A260700. Diaconis and Simper [6] study distributions for specified subgroup pairs, including contingency tables with prescribed margins. Renteln [7] studies Sylow subgroup double quotients. Munarini, Poneti and Rinaldi [8] develop matrix compositions, including fixed-row regimes. These distinctions prevent transferring a neighboring asymptotic result to the present sequence without further work.

[write] Added 3 October 2026, batch 85B (reciprocal note). The matrix compositions of [8] now have their own collection report [13] (OEIS A261780, A261781, A261784). Its route on compact rays uses the same two devices as Sections 3 and 5 here: the Fubini pole at $\rho = \log 2$ is extracted with an exponentially small relative error, and the result is passed through a Stirling transform with nonnegative summands (Munarini, Poneti and Rinaldi's Proposition 29), followed by a saddle point; that report also proves a finite all-orders coefficient algorithm and a Lambert- W index inversion with integer brackets. The sequences, transforms and scales differ (two growing parameters n, k there, with $n!/(2\rho^{n+1})$ and a saddle in n/k), and neither report uses the other's theorems. Question 3 below (exponentially small contributions of the poles $\rho \pm 2\pi i, \dots$) has an open counterpart there, and the exact pole-lattice formula for the Fubini numbers themselves is Theorem `q2:thm:fubini` of [11].

[write] Added 5 October 2026, batch 100 (reciprocal note). The contingency tables of A120733 named in Section 1 now have an unlabelled counterpart in the collection: [14] counts the same nonnegative matrices of total sum n without zero rows or columns modulo separate row and column permutations (OEIS A007716), proves $a_n \sim B_n^2 e^{W(n)^2/2}/n!$ and a complete fixed-order expansion, and uses the weighted count $\sum_{k,l} M_{k,l}(n)/(k!l!)$, whose terms are those of A120733 divided by $k!l!$. No result transfers between the labelled and the unlabelled count, and neither report uses the other.

[write] Added 7 October 2026, batch 108 (reciprocal note). The same Fubini pole at $\rho = \log 2$ is extracted in the collection's report [15] on symmetric nonnegative integer matrices of total sum n without zero lines (OEIS A138178). There the count is $\frac{1}{2} \sum_{k \geq 0} 2^{-k} f_n(k)$ with $f_n(k)$ a polynomial in k ; a positive coefficient envelope (its Lemma 3.1 and (3.6)) bounds the contribution of all the nonreal poles $\rho \pm 2\pi i, \dots$ at once, at every polynomial degree, and the pole at ρ gives $n!/(2\rho^{n+1})$, the ordered Bell number up to an exponentially small relative error (its Corollary 2.2), combined with an involution saddle; the expansion is in powers of $n^{-1/2}$ (its Theorem 2.1). That report bounds the nonreal poles but does not compute their contributions, so it does not bear on Question 3 below. The sequences, transforms and scales differ, and neither report uses the other's theorems.

Several precise questions remain beyond the theorem proved here.

1. **Effective bounds.** Track explicit constants in the contour estimates, logarithmic cutoffs and

Taylor remainders to obtain certified finite- n bounds and computable integer-threshold brackets. This would turn the present asymptotic inverse result into a usable certified algorithm.

2. **Growth of correction coefficients.** Determine the growth and eventual sign behavior of $B_m(\rho)$ as $m \rightarrow \infty$. The four negative corrections computed here do not settle either question. Such estimates are prerequisites for any justified order growing with n .
3. **Exponentially small contributions.** The Fubini generating function has additional poles at $\rho \pm 2\pi i, \dots$, but tracking their contributions through both defect sums requires new uniform analysis. Their existence alone does not prove a transseries, optimal truncation formula, or exponentially improved error for p_n .
4. **Combinatorial meaning of corrections.** The proof separates singleton restrictions, even-block excesses and nontransposition cycles. A direct combinatorial or probabilistic explanation for the cancellations producing the coefficients could clarify the negative first correction and suggest extensions to other Coxeter types.
5. **Other parabolic families.** An analogous all-orders theorem for distinct parabolic coset subsets in another Coxeter family first requires the correct duplicate-identification enumeration. Results for prescribed subgroup pairs cannot substitute for that input.

None of these further questions is claimed solved by the present calculations.

A Third and fourth correction polynomials

The finite exact algorithm gives

$$\begin{aligned}
 B_3(r) &= \frac{r^4}{51840} (135r^{11} - 1485r^{10} + 2745r^9 + 21335r^8 \\
 &\quad - 73620r^7 - 58680r^6 + 318600r^5 - 17172r^4 \\
 &\quad - 203040r^3 + 2160r^2 - 129600r + 47520), \\
 B_4(r) &= \frac{r^4}{2488320} (405r^{16} - 5940r^{15} + 19710r^{14} + 110220r^{13} \\
 &\quad - 791515r^{12} + 281640r^{11} + 7063176r^{10} \\
 &\quad - 10232784r^9 - 17774352r^8 + 28817856r^7 \\
 &\quad + 4124736r^6 + 16329600r^5 + 969408r^4 \\
 &\quad - 38776320r^3 + 6531840r^2 - 6220800r + 2280960).
 \end{aligned}$$

The JSON coefficient file stores rational expressions for $D_0, \dots, D_4$, $U_0, \dots, U_4$ and $B_0, \dots, B_4$, together with decimal evaluations. The symbolic expressions, rather than the rounded evaluations, are authoritative.

B Reproducing the computations

The companion ZIP is self-contained for computational replay after installing Python 3, SymPy and mpmath. It contains this editable L^AT_EX source, the PDF, original programs, reference numerical

data with source attribution, generated exact data, and a SHA-256 manifest. It contains no full third-party papers. From the extracted package root, run:

```
python -m pip install -r requirements.txt
python code/replay.py --quick
python code/replay.py --full
```

The quick replay checks exact enumeration through $n = 40$ and recomputes coefficients through order four; the full replay recomputes all values through $n = 400$. Both enumerate the actual coset subsets through $n = 5$, check the cycle identity, verify exact coefficient expressions, and run inverse diagnostics. Generated replay outputs go to a separate directory and are compared with the supplied data. To request another fixed coefficient order, run:

```
python code/coefficients.py 5 --output-dir replay-order5
```

Runtime and expression size grow with the requested order. There is no claim that arbitrary orders have a bounded practical runtime. To rebuild the article, use a standard L^AT_EX installation with the listed common packages:

```
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

Numerical checks do not determine rigorous finite-input values for the constants in $O_M(\cdot)$. The machine-readable results expressly label the inverse examples as diagnostics.

[write] **Added 2 October 2026, batch 77.** The companion ZIP described above is not shipped as such. In the collection the five programs and `build_pdf.sh` are in `code/`, and the data, `requirements.txt` and the two recorded result files (`results/validation.json` and `results/inverse-validation.json`, now `data/results-validation.json` and `data/results-inverse-validation.json`) in `data/`, all byte-identical to the delivery. The delivered PDF and the SHA-256 manifest `MANIFEST.json` (a checksum ledger, verified at intake and retired) are not shipped; both survive in the archive. Because `replay.py` begins by running `verify_manifest.py`, which reads `MANIFEST.json`, the replay command above stops at its first step in the collection layout. The report's `README.md` gives a rerun of the three computational steps (coefficients, exact values with coset enumeration, inverse diagnostics) that works with the shipped names and writes only to a scratch directory, and the retrieval of the delivered layout for the full replay. `build_pdf.sh` compiles an `article.tex` in its own directory, so in the collection it must be copied next to the article before use.

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